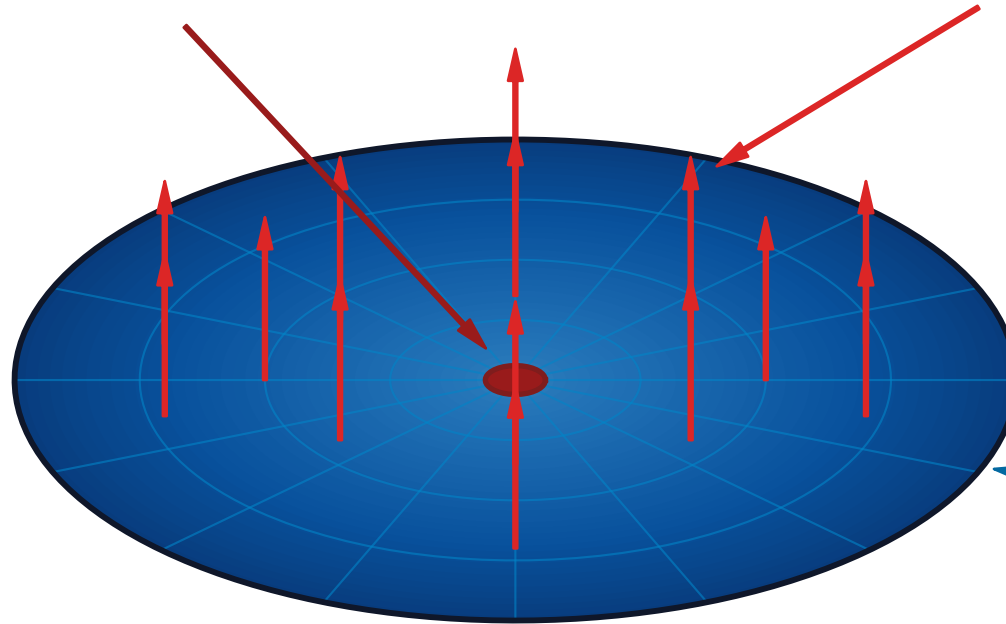


Central Payload Hub Package

- **Hub Radius:** $r_{\text{hub}} = 0.10 \text{ m}$ ($r_{\text{hub}}/R = 0.05$)
- **Payload Mass:** $m_{\text{payload}} = 2.0 \text{ g}$
- **Rigid Payload Instrument Interface**

Uniform Radiation Pressure Load

- **Laser Power:** $P_0 = 100.0 \text{ GW}$
- **Normal Thrust:** $F_0 = 666.40 \text{ N}$
- **Applied Pressure:** $p_{\text{avg}} = 53.09 \text{ Pa}$
- **Normal Vector:** $+\hat{z}$ (Transverse Thrust)



Clamped Tensioning Rim Boundary

- **Outer Radius:** $R = 2.00 \text{ m}$ ($D = 4.00 \text{ m}$)
- **Boundary:** $UX = UY = UZ = 0$
- **Rotations:** $ROTX = ROTY = ROTZ = 0$
- **Composite:** $\text{Si}_3\text{N}_4 + \text{TiO}_2/\text{SiO}_2$ ($13.49 \mu\text{m}$)

Outer Diameter: $D = 4.00 \text{ m}$ (Radius $R = 2.00 \text{ m}$, Area $A = 12.57 \text{ m}^2$)